Real Time Tracking and Alert System for Laptop through Implementation of GPS, GSM, Motion Sensor and Cloud Services for Anti-theft Purposes.

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***Abstract*— This paper report explains the methodology to track a stolen laptop through the implementation of GPS, GSM, Motion Sensor, and Cloud Services. Unlike other existing laptop tracking techniques available in the market, the methodology mentioned in the paper tracks the laptop even if it is not connected to the internet or is in turn off mode. With the implementation of IoT, the owner will be able to track his or her stolen laptop the moment it makes a small movement and can trigger an alarm that will be embedded in the laptop. The alarm will make noise audible up to 10 meters that will make the thief think twice before carrying the laptop with himself. Meanwhile the owner will also be able to monitor the location of his or her stolen laptop through the mobile application installed on his or her phone by communicating with the GPS and GSM modules embedded in the laptop, through the cloud.**

***Index Terms*— GPS; GSM; laptop; accelerometer; GPS; GSM; cloud computing; tracking; alert; alarm; theft.**

I. INTRODUCTION

Taking India as an example which has a whopping population of 1.3 billion people, around 10 million of the population owns a laptop. In the current world, it's almost impossible to imagine that someone can live without computers or laptop. They have become an electronic device of almost every day use for individuals of every age, and essential in almost all the business dealings that are made nowadays. Every laptop is a storage to vital data and information which is very important to its owner and loss or theft of which can lead to serious loss for its owner. According to National Crime Records Bureau, around 1.3 lakh laptops are stolen in the year 2016 and surprisingly the recovery rate was as low as 8%. It is the duty of police officials or crime branch to handle such matters but they always have graver crime to focus upon. It's tough and a tedious job for the officials to track the laptop. Hence, not all laptops get tracked. Crime department tries to track the laptop by hook or crook if the laptop belongs to someone of importance or if it contains vital or crucial data of national importance or of national security. Thus in the end the sufferer is the common man.

Fig 1. A Laptop Model

This research methodology showcases the effective, efficient and economical way through which every man could easily track his or her laptop using GPS module, GSM module, motion sensors and cloud services in case of theft. The whole functioning of the system is entirely based on IoT. A lot software have been developed in the recent years but they are not much effective. They only help the owner of the laptop gets is to track the device only when they discover that

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their device has been stolen and that can only happen when there is a working internet connection. The real time laptop tracking and alert system notifies the user as soon as any fidgeting is done with the laptop. Any small movement of the laptop will alert the owner at that very moment. The owner of the laptop will be able to keep a track of his or her laptop through a mobile application installed on their phone. Owners will also have the privilege to set off a high intensity alarm in the laptop which might make the perpetrator think twice before carrying the laptop with himself.

II. RELATED RESEARCH METHODOLOGY

Absolute LoJack is the leader in data protection and theft recovery, with the only security solution that both locates one's missing device and gets it back to the user. Already embedded in devices' firmware, once activated, the patented Persistence *technology* can survive a factory reset or hard wipe, allowing the industry's only Investigations and Recovery Team to locate and retrieve one's stolen laptop, smartphone or tablet. But the disadvantage of this technology is that one can only track the laptop and garner data if the laptop is connected to the internet. Also this system won't be able to track the laptop the moment it is stolen. The user will only come to know about the activity once he or she discovers that the laptop has actually been stolen. A lot of other software has also been made but they possess the same disadvantages.

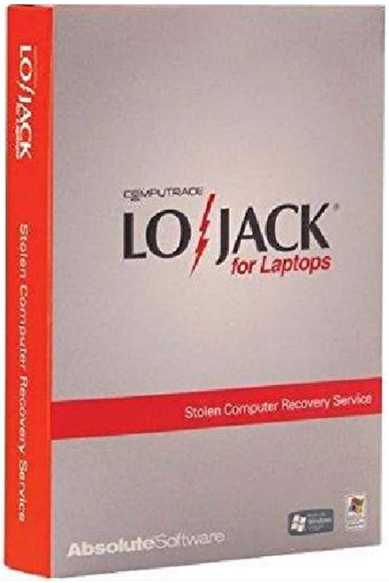

Fig 2. Lojack anti-theft software for laptop security

III. HARDWARE SPECIFICATION

Implementation of the aforementioned product implements the integration of the sensor stack and

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microcontroller and maintaining a stable connection to the cloud sever. While selecting the hardware we were faced with a few restrictions, some physical and some monetary. The product requires that the power consumption and size of the hardware be minimal keeping in mind the overall cost of the device. Due to restrictions in size of the device a small battery had to be selected, in turn this left us with little head room for power consumption. To keep wattage to a bare minimum we selected the STM32 microcontroller,

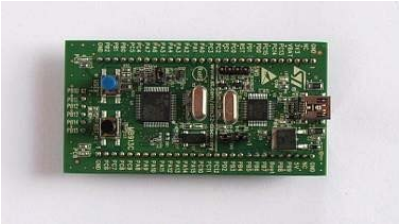

Fig 3. STM32 microcontroller

with dedicated sleep and deep sleep modes this would reduce the overall power consumption during both on and sleep times. The microcontroller will be responsible for receiving data from the sensors processing it and forwarding it to the server over GPRS, in addition to this the MCU will also be activating the anti-theft alarm. A miniature alarm module will be installed to the setup that will go off if the owner thinks there is a security threat to his or her device. The sensor stack incudes a 3-axis accelerometer and a GPS chip. The Bosch Sensortec BMA280 will be used as the accelerometer as it is capable of measuring motion changes in all three axes (X, Y, Z), requiring a supply of 3.6 volts this can easily be powered by the battery. For communication with the MCU we shall be using I2C through the SDA and SCL pins. The GPS chip being used is the Broadcom BCM4775X. All chips being used in the product are SMD in nature allowing for a smaller footprint, furthermore these chips are compliant with our estimated cost targets. The use of flex PCBs will allow for use of the same device in a multitude of laptops as the flex will easily wrap bend around edges and corners and will be less prone to physical damage. A micro USB port will also be provided for easy firmware upgrades to the device, for reflashing firmware a FTDI RS232 USB to TTL converter in series with the USB interface. As for connectivity a SIM900 GSM chip will be used for GPRS

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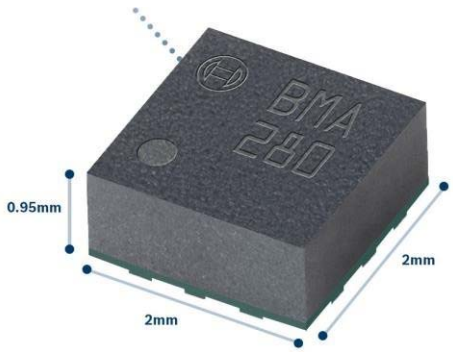

Fig. 4. Bosch Sensortec BMA280 which will be used as the accelerometer

connectivity, the AT command set will be used over a serial connection between the SIM900 chip and the STM32 to send and receive data to and from the server. Another consideration is that different baud

Fig 5. SIM900 GSM chip

rates must be setup for different serial communications as interference will cause malfunctioning of the device. Due to the limitations of the external battery the SIM900 chip will be powered by the laptop's internal battery. The battery being used to power the GPS chip and the MCU has a rating of 0.5 AH(ampere-hour) with inbuilt short circuit and over voltage protection. The battery will be recharged through the internal connections to the USB ports having a rated output of 5V with a maximum current draw of 0.5A.

IV. SOFTWARE SPECIFICATION

The software aspects of the product include integration of the sensor stack, MCU and the Smartphone Application harnessing the Cloud. A web server handles data exchange between the device affixed to the laptop and the smartphone application. The web server exchanges data to and from the device using HTTP GET and POST requests. This data primarily latitude, longitude and a few conditional bits is processed by the server as necessary and the forwarded to the application in JSON format every few seconds. The software on the smartphone app will be designed in such a way that if the laptops sensor's coordinates are outside the bounds of the user or the laptop's position is changed whilst not in the proximity of the user, the user will be immediately informed on the smartphone application and will automatically trigger the alarm on the laptop forcing the thief to drop the item and leave. The end user will

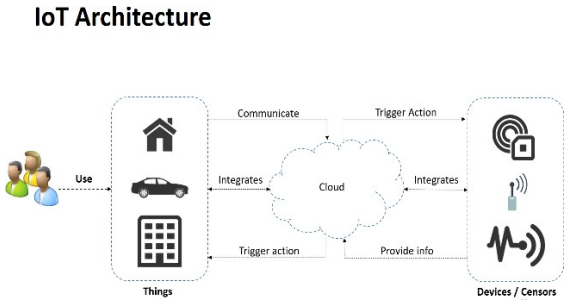

Fig 6. Basic IoT Architecture

have the option to remotely trigger the anti-theft alarm if he/she feel that their laptop's security has been compromised in any way or the laptop has been stolen. The user will also be able to deactivate the alarm if it was triggered erroneously. A condition may arise where the user's phone does not have an active data connection, in such a situation the device will send an SMS to the user's phone number with the Geo-Location coordinates, the user will in turn be able to turn on the alarm through a predefined set of SMS codes.

V. METHODOLOGY

Modern day methods to track stolen laptop using software are not effective. The reason is the owner of the laptop gets to know that their device has been stolen only when they themselves discover that their laptop has been stolen. Furthermore the laptop will be only able to be tracked when it gets connected to the internet. Hence tracking the stolen laptop becomes a long and tedious process. Real Time Tracking and Alert System for Laptop through Implementation of GPS, GSM, Motion Sensor and Cloud Services for Anti-theft Purposes tries to track the stolen laptop by eliminating all those disadvantages that is prevalent in modern day laptop tracking software First of all, the laptop tracking and alarm module that is embedded in the laptop will be connected to a mobile application the accelerometer as it is capable of measuring motion changes in all three axes (X, Y, Z). Now if somebody conspires to steal the laptop, conspirer would definitely try to pick it up, the laptop makes the movement and thus in the process the accelerometer is triggered. For connectivity a SIM900 GSM chip will

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Fig 7. Screenshot of the mobile application for tracking the laptop of the owner.

be used for GPRS connectivity, the AT command set will be used over a serial connection between the SIM900 chip and the STM32 to send and receive data to and from the server. Thus when an accelerometer is triggered, notification is sent to the owner's mobile application about the development. Now if the owner thinks that his or her laptop is under threat, using the mobile application he can trigger the alarm module in the laptop. The alarm will make loud enough noise to be audible up to 10 meters. Thus it will be very tricky and tough situation for the thief or any intruder to carry the laptop with him or her with the alarm system activated as it will alert the passer byes. Now doubts will arise that what will happen if the owner of the laptop himself triggers the alarm when he tries to handle his or her laptop. Thus triggering the alarm will be in the hand of the owner of the laptop as he or she will be controlling the activation and deactivation of alarm remotely through the mobile application on his or her phone or tablet. Meanwhile the owner of the laptop will also be able to keep a constant track of the location or whereabouts of his laptop through the GPS chip embedded in the laptop. The GPS chip being

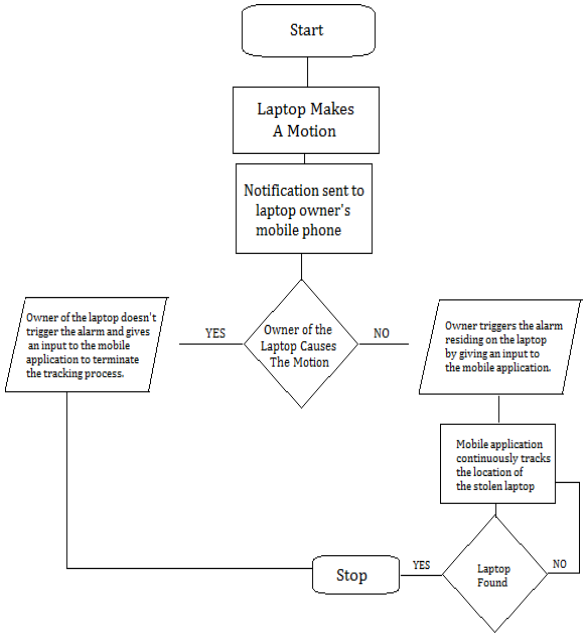

Fig 8. Flowchart of the working of the tracking and alert system

used is the Broadcom BCM4775X. A web server handles data exchange between the device affixed to the laptop and the smartphone application. The web server exchanges data to and from the device using HTTP GET and POST requests. This data primarily latitude, longitude and a few conditional bits is processed by the server as necessary and the forwarded to the application in JSON format every few seconds. The software on the smartphone app will be designed in such a way that if the laptops sensor's coordinates are outside the bounds of the user or the laptop's position is changed whilst not in the proximity of the user, the user will be immediately informed along with the location of the stolen laptop.

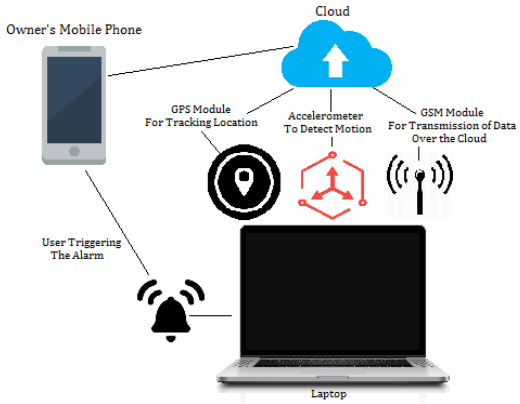

Fig 9. Basic layout of the tracking and alert system

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VI. CONCLUSION

Laptops are always an important asset for its owner. It contains important and vital data and information of its owner. It becomes a huge problem for the owner if his or her laptop gets stolen. Laptop tracking techniques present in the current market are inefficient and not worth the money, since they only are able to track the laptop if it is switched on and is connected to the internet. The methodology mentioned in the paper is an efficient way to track the laptop since it notifies the owner the moment anyone fiddles with the laptop. Also the laptop will be continuously tracked even if it is switched off or not connected to the internet.

VII. FUTURE SCOPE

The current undertakings of the research have been implemented to showcase the real time alert and tracking of stolen laptops. To make the concept more enhanced, the future scope of the project will deal with the ability of the laptop to connect automatically to the internet that will be served by the GSM module. Now the moment the thief opens the laptop and switches it on, a photo will be clicked of the thief through the web cam. Now the internet service will enable the laptop to send the photo which has been clicked to the cloud for the owner of the laptop to see it. The will make the tracking process of the stolen laptop more efficient and easier.

VIII. ACKNOWLEDGMENT

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